

INMO – 2024

Time – 4 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carry equal marks. Maximum marks: 102.
 - Answer to each question should start on a new page. Clearly indicate the question number.
1. In triangle ABC with $CA = CB$, point E lies on the circumcircle of ABC such that $\angle ECB = 90^\circ$. The line through E parallel to CB intersects CA in F and AB in G. Prove that the centre of the circumcircle of triangle EGB lies on the circumcircle of triangle ECF.
 2. All the squares of a 2024×2024 board are coloured white. In one move, Mohit can select one row or column whose every square is white, choose exactly 1000 squares in this row or column, and colour all of them red. Find the maximum number of squares that Mohit can colour red in a finite number of moves.
 3. Let p be an odd prime number and a, b, c be integers so that the integers $a^{2023} + b^{2023}$, $b^{2024} + c^{2024}$, $c^{2025} + a^{2025}$ are all divisible by p divides each of a, b, c .
 4. A finite set S of positive integers is called cardinal if S contains the integer $|S|$, where $|S|$ denotes the number of distinct elements in S . Let f be a function from the set of positive integers to itself, such that for any cardinal set S , the set $f(S)$ is also cardinal. Here $f(S)$ denotes the set of all integers that can be expressed as $f(a)$ for some a in S . Find all possible values of $f(2024)$.
Note: As an example, $\{1, 3, 5\}$ is a cardinal set because it has exactly 3 distinct elements, and the set contains 3.
 5. Let points A_1, A_2, A_3 lie on the circle Γ in counter-clockwise order, and let P be a point in the same plane. For $i \in \{1, 2, 3\}$, let τ_i denote the counter-clockwise rotation of the plane centred at A_i , where the angle of the rotation is equal to the angle at vertex A_i in $\triangle A_1 A_2 A_3$. Further, define P_i to be the point $\tau_{i+2}(\tau_i(\tau_{i+1}(P)))$, where indices are taken modulo 3 (i.e., $\tau_4 = \tau_1$ and $\tau_5 = \tau_2$). Prove that the radius of the circumcircle of $\triangle P_1 P_2 P_3$ is at most the radius of Γ .
 6. For each positive integer $n \geq 3$, define A_n and B_n as

$$A_n = \sqrt{n^2 + 1} + \sqrt{n^2 + 3} + \dots + \sqrt{n^2 + 2n - 1} \text{ and } B_n = \sqrt{n^2 + 2} + \sqrt{n^2 + 4} + \dots + \sqrt{n^2 + 2n}$$
Determine all positive integer $n \geq 3$ for which $\lfloor A_n \rfloor = \lfloor B_n \rfloor$.
Note: For any real number x , $\lfloor x \rfloor$ denotes the largest integer N such that $N \leq x$.

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